

XIWEN CHEN

Agentic AI | Data-Centric Learning | Healthcare Optimization
xiwenc@usc.edu | [GitHub](#) | [LinkedIn](#) | [Personal website](#) | Open to relocate

EDUCATION

University of Southern California | Los Angeles, CA

M.S. in Computer Science, Artificial Intelligence Track; GPA: 3.71 / 4.00 | 2024-01 - 2026-05

South China Normal University | Guangzhou, China

B.E. in Software Engineering; GPA: 3.87 / 4.00 | 2019-09 - 2023-07

EXPERIENCE

Lab for ML, Health and Biomedicine, USC | Los Angeles, CA

Research Lead - Agentic AI, Optimization & Evaluation | 2026-05 - Present

Advisor: Prof. Ruishan Liu

- Lead RadOncGym, a Gymnasium-based environment in which agents tune 12 planning-objective weights and allocate a limited budget of full Gurobi solves using fast surrogate feedback.
- Built standardized evaluation interfaces, automated experiment orchestration, and reproducible trajectory logs; evaluated nine configurations combining three memory mechanisms and three agent role structures.
- Evaluated 600 patient-prediction contexts. Under a 16-turn, three-solve protocol, the best evaluated agent reduced mean clinical-criterion violation by 0.613 Gy versus all-one weights, remaining competitive with 16-solve numerical baselines at 75.0-77.6% less runtime.

2 Cube Global | Los Angeles, CA

AI Engineer Intern - Speech AI & ML Data Pipelines | 2026-01 - 2026-05

- Prototyped multilingual notification TTS pipelines, adapting F5-TTS and comparing it with zero-shot Qwen3-TTS to evaluate quality-cost trade-offs.
- Built Python/Librosa pipelines for 80+ hours of paired speech and transcripts and 45K aligned utterances, filtering 16% misaligned samples for adaptation and held-out evaluation.

University of Southern California | Los Angeles, CA

Research Assistant - Data-Centric AI & Training Data Curation | 2024-09 - 2026-05

Advisor: Prof. Ruishan Liu

- Led first-author TAVO and TAC projects on source and source-target data curation under label scarcity, constrained training budgets, and cross-domain distribution shift.
- Built reproducible PyTorch/CUDA experiments on BraTS 2021, MAMA-MIA, and OfficeHome, including multi-budget ablations and EfficientViT, nnU-Net, and ResNet-50 evaluation.
- At a 150-case source budget, TAVO improved target-averaged Dice over the strongest fixed selectors by 1.0/1.9 points and budget-matched adaptation baselines by 1.0/2.2 points on BraTS/MAMA-MIA.

Xiaomi Technology Co., Ltd. | Beijing & Shenzhen, China

Research Engineer Intern - Distributed Systems & Cloud Infrastructure | 2024-05 - 2024-09

- Integrated network- and erasure-coding components into NCDSS; validated K=4, M=2 storage across a 60-node Linux environment and 50+ client-I/O correctness scenarios.
- Benchmarked eight coding baselines across 48 paired tests and 2 KB-1 GB workloads, improving encoding throughput by 81.5%, upload throughput by 22%, and recovery latency by 12%.

SELECTED MANUSCRIPTS

RadOncGym: An Agent-Based Framework for Radiotherapy Dose-Mimicking Optimization

First author. Manuscript under review, WACV 2027.

TAVO: Target-Aware Source Curation for Budgeted Cross-Center Tumor Segmentation

First author. Manuscript under review.

Target-Anchored Coverage for Source-Target Curation in Cross-Center Medical Image Segmentation

First author. Manuscript under review, NeurIPS 2026.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, SQL, Bash, JavaScript

AI/ML & Evaluation: PyTorch, MONAI, Hugging Face Transformers, scikit-learn, Gymnasium, Gurobi, SGLang, LangGraph

Systems & Infrastructure: Linux, Docker, Git, CUDA, Slurm, Spark, REST APIs, distributed systems, performance benchmarking